

TSYP14 IEEE RAS × IEEE AESS Tunisia Section Chapters Technical Challenge

The Living Map: Spatial Memory for Emergency Robots

Technical Report

1 Introduction & System Concept

1.1 Problem

Robots operating in environments where GPS is unavailable may lose communication or malfunction before returning, resulting in the loss of valuable information that could have contributed significantly to their mission or even saved lives.

The challenge addresses this issue by creating an outside network area to connect the GPS denied area to the global network, and by using a two-robot system where a writer robot creates persistent spatial information and records it in radio beacons, while an executor robot later reconstructs and acts upon that information.

Our approach extends this concept through **multiple** autonomous writer and executor robots whose roles are adapted to the mission requirements. Two writer strategies are considered: a **fast writer** prioritizing rapid victim detection, and a **slower writer** dedicated primarily to systematic map construction while retaining victim-detection capabilities. Victim detection and localization combine **radio-frequency sensing** and **computer vision** with **AI-based processing**, while **simultaneous localization and mapping** provide the robots with their position within the local environment. The robots create a **mesh like beacon network** fully connecting the hazardous area internally, it does so either when last beacon connectivity starts to drop significantly or when there is a relevant event, creating persistent references that can later be used by other robots. Furthermore, **data collected from each mission** is **treated** by the outside network area and sent to command post to be used again in future missions in order to minimize mission failure.

1.2 Selected Environment

Our team chose the **Mines / Tunnels** because in practical terms it is almost always denied of outside world connection because of thick layers of rock or the Faraday Cage physical phenomenon. Furthermore, it is a deeply challenging environment that could be extremely hazardous due to poisonous gas, which makes the use of autonomous exploring and executing robots extremely critical.

2 Technical Solution

2.1 Writer Robots

- **Multiple autonomous Writers** explore different regions of the mine.
- **Slow Writer** : Ground-based mapping robot, prioritizes systematic exploration and map construction, deploys and maintains beacon infrastructure, Performs victim detection when encountered using computer vision and RF sensing.
- **Fast Writer** :Prioritizes rapid event/victim detection ,can rapidly inspect regions inaccessible or inefficient for ground exploration, uses computer vision and RF sensing.
- Both Writer types:
 - Localize themselves using onboard sensors and SLAM.
 - Can detect two events types:
 - * hazardous area encounter.

- * victims encounter.
- Store event information in the beacon network.
- Operate without direct communication with the Outside Network.

2.2 Autonomous Exploration

- The environment is initially unknown; Writers do not require a predefined map.
- **SLAM** continuously updates the local map and robot pose.
- **Frontier-based exploration** identifies boundaries between explored and unexplored regions.
- A **modified Depth-First Search (DFS) strategy** systematically explores tunnel branches.
- A **utility function** prioritizes available frontiers according to factors such as: Expected information gain, distance, traversal energy cost, accessibility, mission relevance...
- Exploration loop:

Sense → Localize → Map → Identify Frontiers → Evaluate → Select → Navigate

2.3 Spatial Memory and Beacon Network

- Beacons create a **persistent spatial memory** inside the mine.
- Each beacon stores relevant information including: event identifier, event type, position, timestamp, detection confidence, source/Writer identifier.
- Writers progressively deploy beacons during exploration.
- **ESP-NOW** is used for communication between Writers and beacons and between beacons.
- Beacons forward relevant information through the network.
- Previously stored information remains available even if the Writer that generated it fails.
- Beacon spacing is determined primarily by the communication requirements of the ESP-NOW network.

2.4 Frame Translation

- Writers operate without GPS inside the mine and therefore estimate their position using **SLAM and onboard sensor fusion**.
- The Writer's position is initially expressed in its local SLAM/map coordinate frame.
- Each deployed beacon stores the Writer's position relative to the beacon network.
- The first beacon of an entrance has a known position in the global reference frame.
- The beacon chain therefore provides the reference needed to transform the Writer's local coordinates into global coordinates.
- This transformation allows detected events to be associated with the global GPS coordinates required by the mission.
- The same reference can subsequently be used by the Executor to navigate from the known entrance toward the stored target.

2.5 Victim Detection

Computer Vision

- Cameras provide direct visual victim detection.
- Detected victims are associated with the Writer's estimated SLAM position.
- Detection confidence is stored with the event.

Wi-Fi CSI

- Wi-Fi Channel State Information is investigated as a complementary sensing modality.
- **Phase 1 objective: Level-1 human-presence detection.**
- Multiple spatially distributed CSI nodes observe changes in the wireless channel caused by human presence.
- CSI measurements are processed and classified using AI.

- RF detection is associated with the Writer’s current spatial region.
- Phase 2 can extend the system toward more precise RF-based localization.

2.6 Distributed Wi-Fi CSI Infrastructure

- Wi-Fi is **generated inside the mine by the robotic/beacon infrastructure**, rather than relying on external connectivity.
- Writers and/or beacons can act as Wi-Fi sources/access points for the CSI sensing network.
- Dedicated **ESP32-S3 CSI nodes** are distributed around the monitored region.
- CSI sensing infrastructure consists of: Wi-Fi source(s), multiple CSI sensing nodes, CSI aggregation/processing, AI-based presence classification.
- The CSI infrastructure is distributed independently from the ESP-NOW communication topology.
- **ESP-NOW**: robot/beacon communication.
- **Wi-Fi CSI**: RF victim-presence sensing.
- Sensing-node density can therefore be increased locally without requiring the entire beacon network to have the same density.

2.7 Victim Detection Fusion

- Computer vision and CSI provide independent evidence of human presence.
- Each detection is associated with: event type, spatial region, timestamp, confidence, detection source.
- Multiple observations can be combined into a single **victim hypothesis**.
- Phase 1 focuses on detecting and recording the presence of a possible victim.
- Phase 2 can use multiple CSI links and additional sensing to estimate the victim’s position more precisely.

2.8 Data Flow

- Sensors generate environmental and robot-state information.
- SLAM converts sensor measurements into a spatial estimate.
- Vision and CSI generate event observations.
- Event observations are converted into compact spatial event records.
- Event records are transmitted through the beacon network using ESP-NOW.
- The Outside Network receives information forwarded from the beacon infrastructure.
- The Command Post uses accumulated information to prepare Executor missions.
- The Executor retrieves relevant information through the beacon network and navigates toward the target.

2.9 Outside Network Area

- Consists of one or more small stations positioned outside the restricted Inside Area.
- Each station can include: communication computer, directional antennas/parabolic antennas, solar panels, high-capacity battery, optical-fibre cable.
- The optical-fibre cable connects the station to the **first beacon of each mine entrance**.
- Provides the communication bridge between the internal beacon network and the Command Post.
- Receives mission information from the Inside Area and forwards it to the Command Post.
- Receives mission commands and relevant information from the Command Post and transfers them toward the appropriate entrance.
- The stations provide a local energy source for **robot recharging** between missions.
- Command Posts exchange mission information with a **global server**, allowing knowledge from previous missions to contribute to the planning and analysis of future missions.

2.10 Executor

- Executor receives a mission based on previously collected information.
- It does not require the Writer that collected the information to remain operational.

- Executor uses the beacon network for: localization, route reconstruction, target identification.
- Executor can be deployed once sufficient information exists, without requiring complete exploration of the mine.

3 Failure Cases and Resilience

- **Writer failure:** Previously transmitted information remains stored in beacons. / Another Writer or Executor can continue from the preserved information.
- **Beacon failure :** Additional beacons can extend or restore the communication chain. / Previously stored information remains available in neighbouring nodes.
- **Incomplete exploration :** Explored regions and detected events remain available even if the Writer cannot complete its route.

4 Implementation and Validation Plan

Stage 1 — Simulation

- Model mine entrances, branches, obstacles and event locations.
- Validate autonomous exploration and mission allocation.

Stage 2 — SLAM and Autonomy

- Implement sensor fusion and SLAM.
- Validate frontier detection and modified DFS.
- Validate utility-based frontier selection.

Stage 3 — Beacon Network

- Implement ESP-NOW communication.
- Measure: communication range, packet reliability, beacon spacing, forwarding reliability.

Stage 4 — Spatial Memory

- Implement event records.
- Validate storage and retrieval of: event, position, timestamp, confidence, source.

Stage 5 — Victim Detection

- Integrate computer vision.
- Develop the CSI sensing prototype.
- Test Level-1 human-presence detection.

Stage 6 — Integrated System

- Combine: autonomy, SLAM, beacon network, computer vision, CSI sensing.
- Validate event transmission and persistence.

Stage 7 — Executor

- Generate missions from stored information.
- Validate beacon-based navigation toward detected targets.

Stage 8 — Failure Testing

- Writer failure.
- Beacon failure.
- Communication degradation.
- Incomplete exploration.
- Uncertain detection.
- Partial map availability.

5 References

- [1] Espressif Systems, “ESP-NOW,” *ESP-IDF Programming Guide*.
- [2] S. et al., “WiFi Sensing Model for Intrusion Detection in Smart Home Environment,” in *Proc. 2023 IEEE 11th Region 10 Humanitarian Technology Conference (R10-HTC)*, Rajkot, India, 2023, doi: 10.1109/R10-HTC57504.2023.10461789.
- [3] “Wi-CaL: WiFi Sensing and Machine Learning Based Device-Free Crowd Counting and Localization,” *IEEE Access*, vol. 10, pp. 24395–24410, 2022, doi: 10.1109/ACCESS.2022.3155812.
- [4] “Device-Free Presence Detection and Localization With SVM and CSI Fingerprinting,” *IEEE Sensors Journal*, vol. 17, no. 23, pp. 7990–7999, 2017, doi: 10.1109/JSEN.2017.2762428.